

THOMPSON, MB, Canada

WMO: 710790

Lat: 55.800N

Lon: 97.860W

Elev: 224

StdP: 98.66

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-38.5	-36.1	-42.5	0.1	-38.4	-40.5	0.1	-35.9	8.9	-20.1	8.1	-20.8	1.4	300	0.738

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	27.9	17.9	26.2	17.1	24.6	16.3	19.6	25.3	18.5	23.8	17.4	22.6	4.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.6	13.0	22.1	16.5	12.1	20.7	15.3	11.1	19.7	57.2	25.2	53.4	23.8	49.9	22.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.9	7.0	DB	-42.1	31.7	2.6	1.9	-43.9	33.0	-45.4	34.1	-46.9	35.2	-48.7	36.5	
			WB	-42.1	21.6	2.5	1.3	-43.9	22.6	-45.3	23.3	-46.7	24.0	-48.5	25.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.3	-22.8	-20.1	-12.5	-2.3	6.0	13.2	16.5	14.7	8.7	0.7	-11.0	-19.9
	DBStd	15.20	7.71	7.28	8.10	7.21	5.31	4.34	3.32	3.66	4.59	4.68	7.35	8.25
	HDD10.0	5024	1018	842	699	371	147	17	1	4	79	290	630	927
	HDD18.3	7575	1276	1076	957	619	383	161	77	124	291	547	880	1185
	CDD10.0	529	0	0	0	1	23	114	201	149	39	2	0	0
	CDD18.3	39	0	0	0	0	1	7	19	11	1	0	0	0
	CDH23.3	626	0	0	0	0	22	144	278	158	23	0	0	0
	CDH26.7	113	0	0	0	0	4	26	52	29	2	0	0	0
	WSAvg	3.2	3.0	3.2	3.4	3.6	3.8	3.4	3.1	3.2	3.2	3.3	2.9	2.8
Precipitation	PrecAvg	531	22	17	24	25	44	63	91	88	64	44	30	19
	PrecMax	875	52	46	54	81	101	144	210	257	120	112	102	45
	PrecMin	380	4	3	3	1	7	7	33	23	19	17	9	4
	PrecStd	98	12	8	13	20	23	30	44	54	31	21	20	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.0	1.1	8.8	19.6	25.7	29.5	30.6	29.7	26.2	17.1	5.4	-0.7
		MCWB	-1.6	0.0	4.0	10.0	14.4	18.1	19.6	19.4	17.8	11.2	3.0	-1.5
	2%	DB	-4.8	-1.7	5.8	14.8	22.5	26.8	28.0	27.1	22.7	13.1	2.3	-3.3
		MCWB	-5.5	-2.7	2.1	7.6	12.7	16.3	18.2	18.2	15.5	8.9	1.0	-3.9
	5%	DB	-7.6	-5.2	2.6	11.3	19.6	24.9	26.3	25.1	19.6	10.0	0.5	-5.5
		MCWB	-8.0	-6.1	0.0	5.2	11.0	15.5	17.4	17.4	14.0	7.1	-0.3	-6.0
	10%	DB	-10.7	-8.2	-0.2	8.4	16.5	22.8	24.7	22.6	16.8	7.5	-1.0	-7.4
		MCWB	-11.1	-8.9	-1.7	3.6	9.6	14.6	16.8	16.2	12.5	5.7	-1.8	-7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.7	0.1	4.2	10.8	15.5	20.7	21.2	20.7	18.4	11.9	3.1	-1.4
		MCDB	-1.2	1.2	7.9	18.3	23.6	25.7	27.4	26.8	24.0	15.0	4.6	-0.8
	2%	WB	-5.4	-2.7	2.3	8.0	13.4	18.3	19.7	19.3	16.3	9.4	1.2	-3.9
		MCDB	-4.8	-1.7	5.4	14.3	20.8	23.7	25.5	25.0	21.8	12.7	2.1	-3.4
	5%	WB	-8.0	-6.1	0.3	5.8	11.7	16.7	18.6	18.2	14.6	7.4	-0.2	-5.9
		MCDB	-7.6	-5.3	2.3	10.6	18.5	23.1	24.0	23.1	18.6	9.7	0.5	-5.5
	10%	WB	-11.1	-9.0	-1.8	3.8	10.0	15.2	17.5	17.1	12.9	5.7	-1.8	-7.8
		MCDB	-10.6	-8.1	-0.1	8.0	16.1	21.2	22.8	21.7	16.9	7.5	-1.1	-7.4
Mean Daily Temperature Range	5% DB	MDBR	9.5	12.3	14.0	13.3	13.0	13.3	12.8	12.0	11.1	7.5	8.3	9.0
		MCDBR	9.7	12.2	14.2	16.6	18.8	17.2	16.0	15.8	15.8	12.8	8.4	8.8
	5% WB	MCWBR	9.5	11.2	10.8	10.5	10.5	8.5	8.0	8.4	9.4	8.7	7.3	8.5
		MCDBR	9.6	11.4	12.7	15.5	16.7	14.5	13.5	13.5	13.9	11.5	8.0	8.7
		MCWBR	9.4	10.6	9.8	10.0	9.7	7.7	7.4	7.8	8.9	8.2	7.1	8.5
Clear-Sky Solar Irradiance	taub		0.242	0.241	0.276	0.318	0.334	0.363	0.402	0.359	0.321	0.287	0.250	0.219
	taud		2.193	2.314	2.274	2.344	2.453	2.390	2.329	2.467	2.525	2.572	2.352	2.194
	Ebn at Noon		687	859	899	905	909	881	834	855	847	803	689	632
	Edh at Noon		55	72	97	107	102	111	116	95	77	58	47	42
All-Sky Solar Radiation	RadAvg		0.82	1.92	3.50	5.12	5.58	5.76	5.36	4.32	2.91	1.51	0.72	0.52
	RadStd		0.08	0.14	0.28	0.41	0.48	0.51	0.37	0.36	0.34	0.21	0.07	0.08

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page